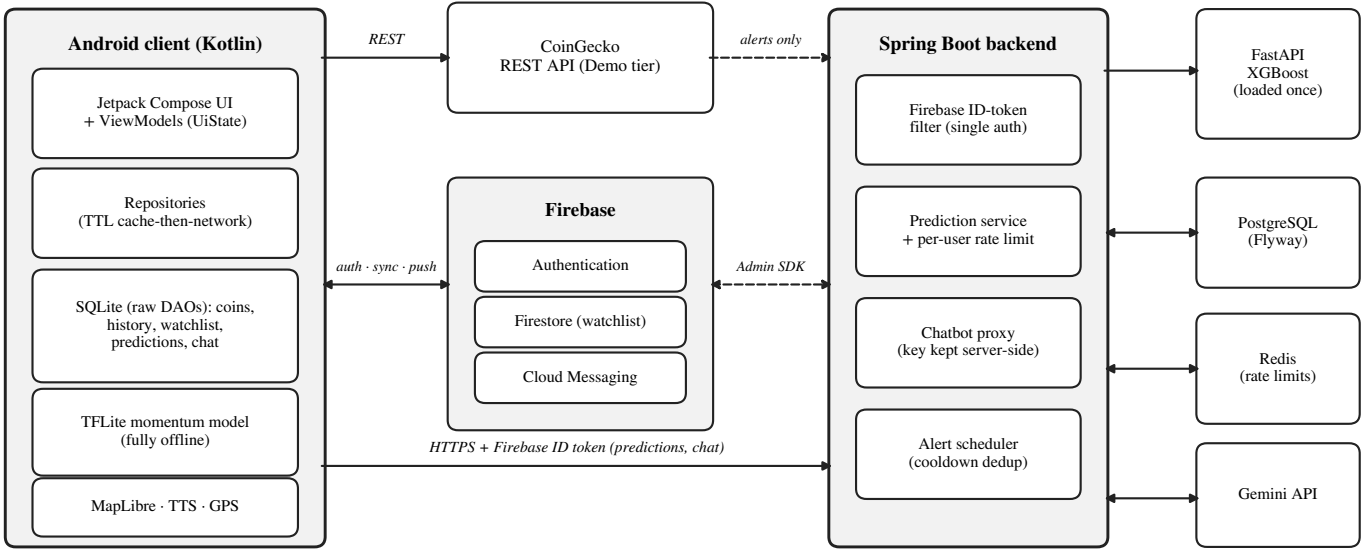
Android client (Kotlin)
Jetpack Compose UI
+ ViewModels (UiState)
Repositories
(TTL cache-then-network)
SQLite (raw DAOs): coins,
history, watchlist,
predictions, chat
TFLite momentum model
(fully offline)
MapLibre · TTS · GPS
REST
CoinGecko
REST API (Demo tier)
alerts only
Firebase
Authentication
Firestore (watchlist)
Cloud Messaging
auth · sync · push
Admin SDK
HTTPS + Firebase ID token (predictions, chat)
Spring Boot backend
Firebase ID-token
filter (single auth)
Prediction service
+ per-user rate limit
Chatbot proxy
(key kept server-side)
Alert scheduler
(cooldown dedup)
FastAPI
XGBoost
(loaded once)
PostgreSQL
(Flyway)
Redis
(rate limits)
Gemini API